

Part	Quantity	Mass per.	Total
Pyro valve	9	0.2	1.8
Latch valve	4	0.32	1.28
Solenoid valve	2	0.2	0.4
Proportional Control Valve	2	0.115	0.23
Pressure transducer	10	0.2	2
Pressure regulator	2	0.45	0.9
Service valve	9	0.209	1.881
Filter	11	0.18	1.98
Orifice	6	0.05	0.3
Temperature sensor	17	-	-
Check valve	10	0.2	2
TOTAL			12.771

IPS

Part	Quantity	Mass per.	Total
Pyro valve	18	0.2	3.6
Latch valve	14	0.32	1.92
Pressure transducer	5	0.2	1
Pressure regulator	2	0.45	0.9
Service valve	5	0.209	1.045
Filter	18	0.18	1.08
Orifice	1	0.05	0.05
Temperature sensor	14	-	
Check valve	10	0.2	2
Level transducer	2	-	

Solenoid valve	4	0.2	0.8
TOTAL			12.395

ACS

Total Feed system mass(w/o tanks and engines) : =
12.395 + 12.771 = 14.395 kg

Thermo sensor (thermocouple specs)

Type	Material Composition	Temperature Range	Standard Limits	Use Cases	Notes
K	Chromel/Alumel	-280°C - 1260°C	± 2.2°	Aerospace, nuclear, automotive	Most commonly used thermocouple
J	Iron/Constantan	0°C - 750°C	± 2.2°	Plastics industry	Use below ambient temperature causes rust
E	Chromel/Constantan	-200°C - 900°C	± 1.7°	Cryogenics, aviation, flow chambers	Not ideal in a vacuum
T	Copper/Constantan	-270°C - 350°C	±1°	Cryogenics, ultralow freezers	Best for low-temperature applications

Tank Specs

OA's P/N	Total Volume (in³)	Total Volume (L)	Qualified Propellant Volume (in³, lbm)	Qualified Propellant Volume (L, kg)	Tank Size (in)	Tank Size (mm)
80274-1	3,660	60	2,748 in³	45.1 L	19.06 ID (ICD for 80274)	484 ID

Weight (lbm)	Weight (kg)	Operating Pressure (psig)	Operating Pressure (bar)	Programs	Note
14.77	6.7	377	26	EXOSAT, VCL, Dawn, OCO, OCO-2, MSL, TESS, Orbview 3	80274 Qualified Environments

IPS

Part	Quantity	Mass per.	Total
Next-C Ion thruster	2	14	28
Xenon Tanks	6	11.1	66.6
TOTAL			94.6

ACS

Part	Quantity	Mass per.	Total
Aerojet Rocketdyne MR111G	12	0.37	4.44
Hydrazine Tanks (Orbital ATK 80274-1)	2	6.7	13.4
TOTAL			17.84

Total Feed system mass : = **14.395 + 17.84 + 94.6 =**
126.835